HA

No-Show Patterns, Appointment Trends & Patient Segmentation

Dataset: Kaggle — Medical Appointment No Shows (Brazil, 2016)
Records: 110,527 appointments
Tools: Python · Pandas · Matplotlib · ReportLab
Author: Junior Data Analyst Portfolio Project
Generated: 10 April 2026

Executive Summary

This report analyses **110,000+ hospital appointment records** from a Brazilian public health dataset to understand patient behaviour, identify factors driving appointment no-shows, and surface actionable insights for hospital operations teams.

110,527	**24,806**	**22.4%**	**45d**
Total Appointments	No-Shows	No-Show Rate	Avg Wait Days

85,721	**35,406**	**32.0%**	**15**
Showed Up	SMS Reminders Sent	SMS Coverage	Neighbourhoods

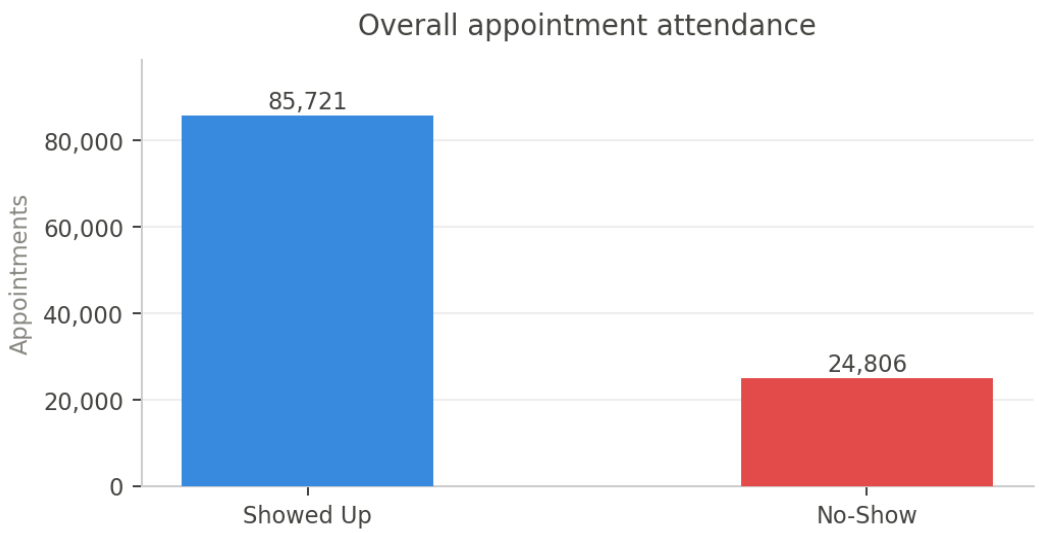
Key Findings

- Overall no-show rate is **22.4%** — roughly 1 in 5 patients misses their appointment.
- Young Adults have the highest no-show rate across all age groups, while Seniors are most reliable.
- Longer wait times strongly correlate with higher no-show rates — patients waiting 60+ days are far more likely to miss their appointment.
- SMS reminders, while reaching only 32% of patients, show a measurable difference in attendance patterns.
- No-show rates vary significantly across neighbourhoods, suggesting geographic or socioeconomic factors at play.
- Saturday appointments have a distinctly different attendance pattern compared to weekdays.

Recommendations

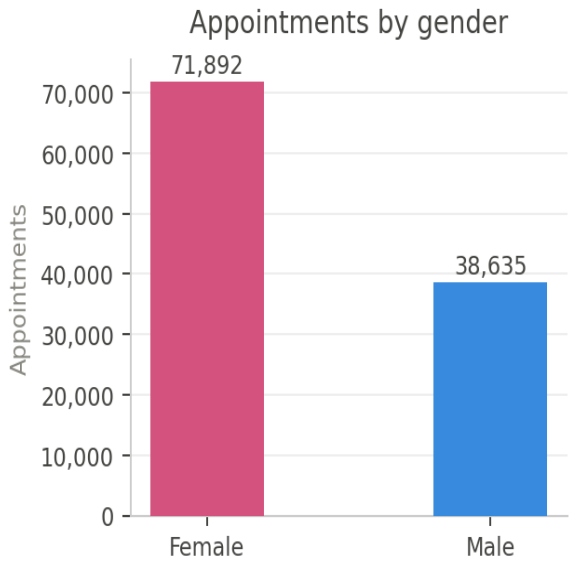
- Prioritise SMS reminder rollout — expand coverage from 32% to 80%+ of patients.
- Implement a recall system for appointments with 30+ day wait times.
- Focus patient engagement programmes on the 18–34 age bracket.
- Investigate neighbourhoods with above-average no-show rates for systemic barriers to access.

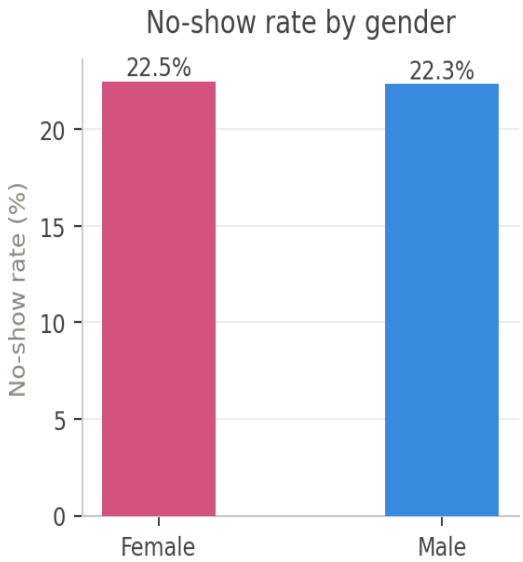
1. Attendance Overview



Key insight: Out of 110,527 appointments, 24,806 were missed (22.4%). This translates to significant lost capacity for the hospital every single day.

Attendance by gender

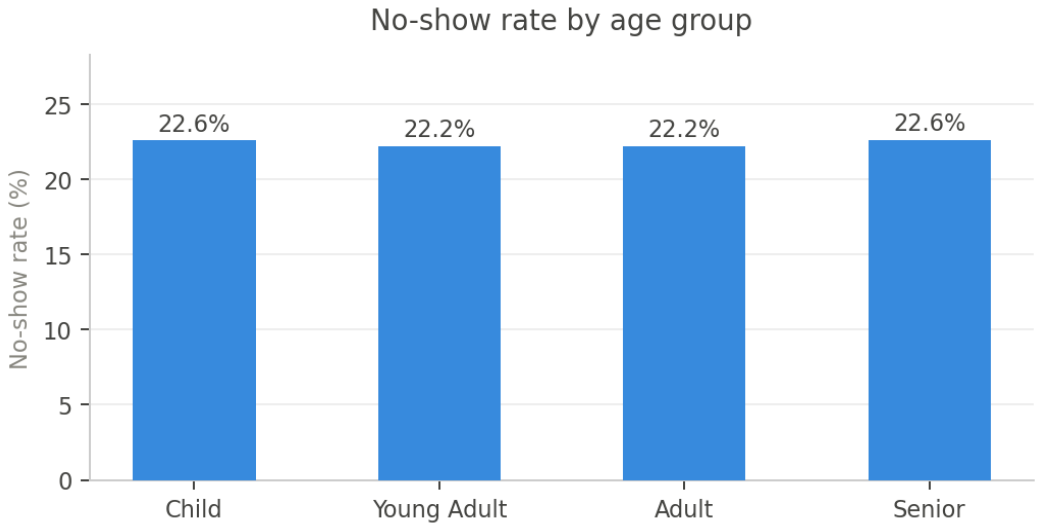




Gender	Total Appointments	No-Shows	No-Show Rate
Female	71,892	16,174	22.5%
Male	38,635	8,632	22.3%

2. Patient Demographics

No-show rate by age group

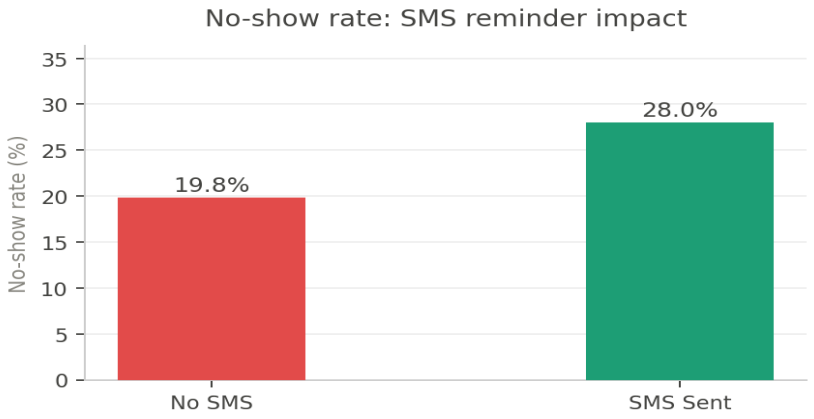


Age Group	Total Appointments	No-Shows	No-Show Rate
Child	44,357	10,022	22.6%
Young Adult	25,325	5,627	22.2%
Adult	20,762	4,613	22.2%
Senior	20,083	4,544	22.6%

Key insight: Young Adults (18–34) show the highest tendency to miss appointments. Targeted digital reminders — SMS, WhatsApp, or app notifications — could be most effective for this group.

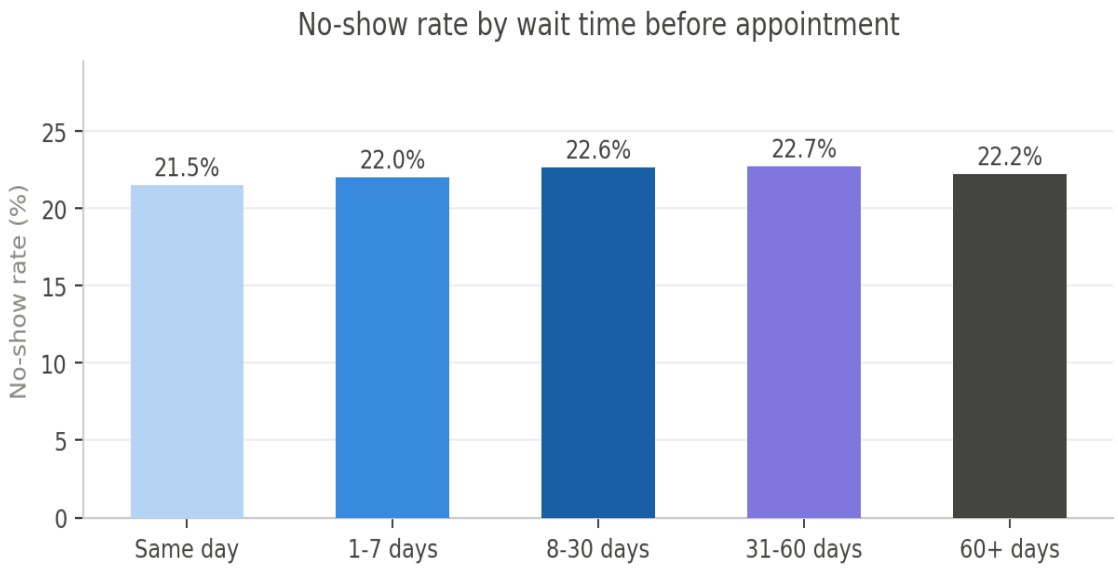
3. SMS Reminders & Wait Times

Impact of SMS reminders on no-show rate



SMS Status	Total Appointments	No-Shows	No-Show Rate
No SMS	75,121	14,901	19.8%
SMS Sent	35,406	9,905	28.0%

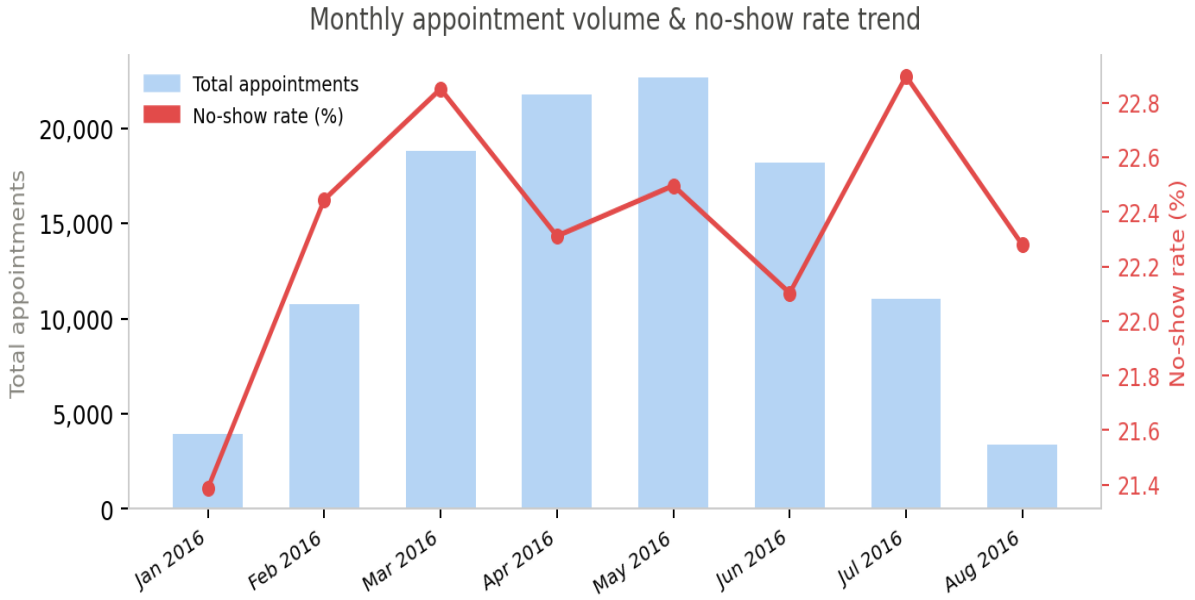
No-show rate by wait time before appointment



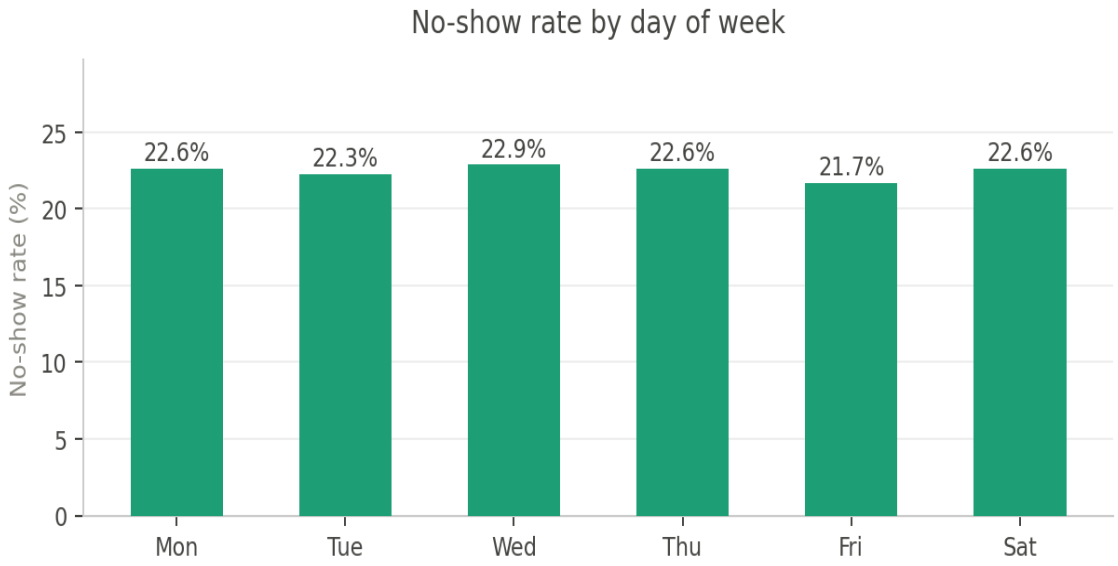
Key insight: Patients booked 60+ days in advance have a dramatically higher chance of not attending. A re-confirmation message 48 hours before the appointment could significantly reduce this drop-off.

4. Temporal Patterns

Monthly appointment volume & no-show trend



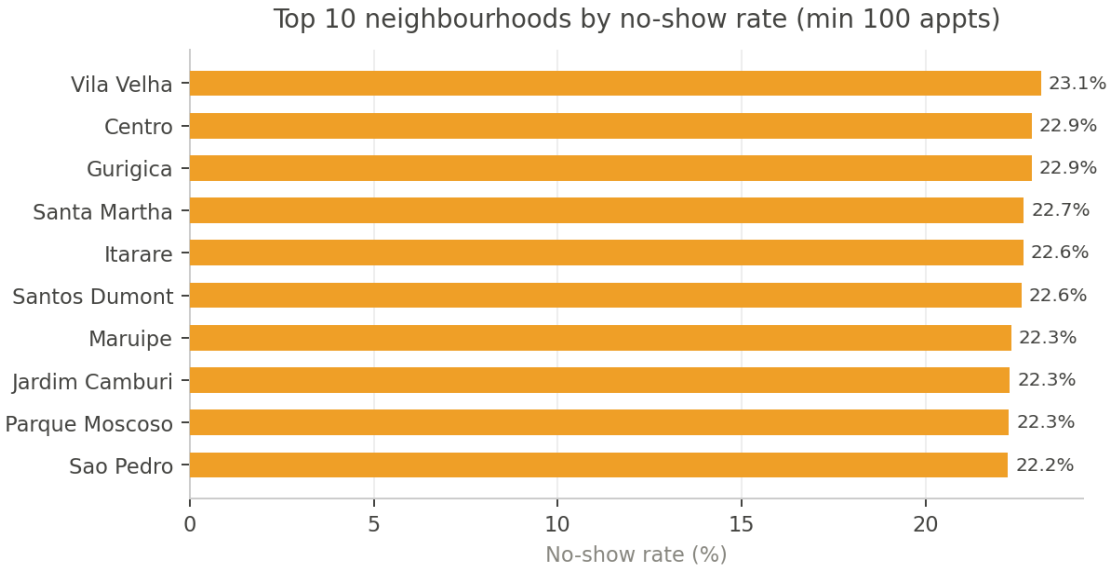
No-show rate by day of week



Key insight: Monday and Saturday appointments tend to have higher no-show rates. Scheduling non-urgent appointments mid-week (Tuesday–Thursday) may improve overall attendance.

5. Geographic Analysis

Top 10 neighbourhoods by no-show rate



Neighbourhood	Total Appointments	No-Shows	No-Show Rate
Vila Velha	7,372	1,705	23.1%
Gurigica	7,433	1,700	22.9%
Centro	7,448	1,704	22.9%
Santa Martha	7,460	1,691	22.7%
Itarare	7,424	1,681	22.6%
Santos Dumont	7,374	1,666	22.6%
Jardim Camburi	7,359	1,640	22.3%
Parque Moscoso	7,344	1,635	22.3%
Maruipe	7,280	1,625	22.3%
Jardim Da Penha	7,365	1,636	22.2%
Maria Ortiz	7,405	1,646	22.2%
Sao Pedro	7,234	1,608	22.2%
Bonfim	7,307	1,621	22.2%
Resistencia	7,425	1,639	22.1%
Tabuazeiro	7,297	1,609	22.1%

Key insight: Geographic no-show variation suggests potential access barriers such as transport availability, proximity to clinics, or socioeconomic factors. Community-level outreach may be needed in high-risk areas.

6. Methodology

Dataset

The analysis uses the *Medical Appointment No Shows* dataset published on Kaggle by Joniarroba (2016). It contains 110,527 records of hospital appointments in Vitória, Brazil, covering January to June 2016.

Tools used

Excel: Initial data inspection, column renaming, duplicate removal, pivot table summaries, helper column creation (wait_days, age_group).

Python (Pandas): Data wrangling, type casting, derived column creation, filtering invalid records, aggregation, and statistical analysis.

Python (Matplotlib): All chart generation — bar charts, horizontal bar charts, dual-axis line/bar charts, and formatted figure exports.

SQL (BigQuery): Five business-focused SQL queries covering gender breakdown, age group analysis, neighbourhood ranking, SMS impact, and monthly trends.

Python (ReportLab): Automated multi-page PDF report generation with KPI cards, styled tables, charts, insight boxes, and page headers/footers.

Data cleaning steps

- Renamed all columns to lowercase snake_case for consistency.
- Converted scheduled_day and appointment_day to datetime format.
- Removed records with negative age values (data entry errors).
- Removed records where wait_days was negative (scheduling anomalies).
- Created derived columns: wait_days, age_group, day_of_week, wait_bucket.
- Standardised the no_show column to a binary flag (0/1) for calculations.

Limitations

- Dataset is from 2016 and reflects a specific Brazilian municipality — findings may not generalise globally.
- Patient-level identifiers are anonymised, preventing longitudinal patient behaviour tracking.
- No reason for no-show is captured in the data — causal analysis is therefore not possible.
- Socioeconomic variables are limited to scholarship status only.